Particulars of the Exercises Performed

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CIE Marks (Reduced to 50) =Total/5

Problem 1:
An electronic company is engaged in the production of two components C_1 and C_2 that are used in radio sets. Each unit of C_1 costs the company Rs. 5 in wages and Rs. 5 in material, while each of C_2 costs the company Rs. 25 in wages and Rs. 15 in material. The company sells both products on one period credit terms, but the company's labour and material expenses must be paid in cash. The selling price of C_1 is Rs. 30 per unit and of C_2 it is Rs. 70 per unit. Because of the company's strong monopoly in these components, it is assumed that the company can sell, at the prevailing prices, as many units as it produces. The company's production capacity is, however, limited by two considerations. First, at the beginning of period 1, the company has an initial balance of Rs. 4,000 (cash plus bank credit plus collections from past credit sales). Second, the company has, in each period, 2,000 hours of machine time and 1,400 hours of assembly time. The production of each C_1 requires 3 hours of machine time and 2 hours of assembly time, whereas the production of each C_2 requires 2 hours of machine time and 3 hours of assembly time. Formulate this problem as an LP model so as to maximize the total profit to the company.

Formulation:
Decision Variables:
Let x_1 be the number of components of C_1 produced respectively
Let x_2 be the number of components of C_2 produced respectively

Objective Function: As the company produces as to maximise the total profit to the company 20rs/unit for C_1 and 30rs/unit for C_2.

Maximise Profit $Z(max) = 20x_1 + 30x_2$

Constraints:

Budget (wages + material)	$10x_1 + 40x_2 \leq 4000$
Machine Time	$3x_1 + 2x_2 \leq 2000$
Assembly Time	$2x_1 + 3x_2 \leq 1400$
Non-negative constraints	$x_1, x_2 \geq 0$

Inference:
On this experiment we understood how a company that makes two products (C_1 and C_2) can plan its production to earn the highest possible profit the company has limited resources like money, machine time and assembly time so it can't produce unlimited quantities. By carefully planning how many units of each product to make company can use its available resource in the best possible way to maximise profit. Max Z = ₹8,000, when $x_1 = 400$, $x_2 = 0$

Problem 2:

A company has 3 operational departments weaving, processing and packing with the capacity to produce 3 different types of clothes that are suiting, shirting and woolen yielding with the profit of Rs. 2, Rs. 4 and Rs. 3 per meters respectively. 1m suiting requires 3mins in weaving 2 mins in processing and 1 min in packing. Similarly, 1m of shirting requires 4 mins in weaving 1 min in processing and 3 mins in packing while 1m of woolen requires 3 mins in each department. In a week total run time of each department is 60, 40 and 80 hours for weaving, processing and packing department respectively. Formulate a LPP to find the product to maximize the profit.

Formulation:

Decision Variables:

Let x_1 be the number of units produced in suiting per metre

Let x_2 be the number of units produced in shirting per metre

Let x_3 be the number of units produced in yielding per metre

Objective Function:

As the company produces and sells in market it gets a profit of ₹2 for sitting, ₹4 for shirting and ₹3 for woolen yielding so the objective function here is maximisation.

Constraints:

$$\text{Max } Z = 2x_1 + 4x_2 + 3x_3$$

weaving for suiting takes 3 mins, 4 mins for shirting and 3 mins for woolen yielding and a total 60 hours in a week

$$3x_1 + 4x_2 + 3x_3 \leq 3600$$

Processing for shirting takes 1 min, 2 mins of suiting and 3 mins for woolen yielding and a total of 40 hours in a week.

$$2x_1 + x_2 + 3x_3 \leq 2400$$

Packing for suiting takes 1 mins, 3 mins for shirting and 3 mins for woolen yielding and a total of 80 hours (4800 mins) in a week.

$$x_1 + 3x_2 + 3x_3 \leq 4800$$

Since length cannot be negative x_1, x_2 and x_3 are greater than or equal to 0.

Inference:

$$x_1, x_2, x_3 \geq 0$$

Max Z, ₹3600

when $x_1 = 0$, $x_2 = 900$, $x_3 = 0$

Problem 3:

A bank is in the process of devising a loan policy that involves a maximum of Rs. 12 million. The following table provides the pertinent data about available types of loans.

Type of Loan	Interest Rate	Bad-debt Ratio
Personal	0.140	0.10
Car	0.130	0.07
Home	0.120	0.03
Farm	0.125	0.05
Commercial	0.100	0.02

Bad-debts are unrecoverable and produce no interest revenue. Competition with other financial institutions requires the bank to allocate at least 40% of the funds to farm and commercial loans. To assist the housing industry in the region, home loans must equal at least 50% of the personal, car and home loans. The bank also has a stated policy of not allowing the overall ratio of bad-debt on all loans not to exceed 4%. Formulate the problem as Linear Programming Problem.

Formulation:

Decision Variables:

Let x_1 be the amount alloted to personal loan

Let x_2 be the amount alloted to car loan

Let x_3 be the amount alloted to home loan

Let x_4 be the amount alloted to farm loan

Let x_5 be the amount alloted to commercial loan

Objective Function:

Bank has alloted maximum of ₹12 million for all loans. 40% loans of the total funds is allocated to farm and commercial loans. 50% of personal, car and home loans is allocated to home loan. Overall ratio of bad-debt on all loans should not exceed.

Constraints:

$$\text{Max } Z = (1-0.1)\,0.14x_1 + (1-0.07)0.13x_2 + (1-0.03)\,0.12x_3 + (1-0.05)\,0.125x_4 + (1-0.02)0.1x_5$$

$$x_1 + x_2 + x_3 + x_4 + x_5 \leq 1,20,00,000$$

Inference:

$$0.4(x_1+x_2+x_3+x_4+x_5) \leq x_4+x_5$$

$$0.5(x_1+x_2+x_3+x_4+x_5) \leq x_1+x_2+x_3$$

$$0.1x_1 + 0.07x_2 + 0.03x_3 + 0.05x_4 + 0.02x_5 \leq 0.04$$

Non-negative constraints – $x_1, x_2, x_3, x_4, x_5 \leq 0.04$

Sl. No	Particulars	Max Marks	Marks Obtained
1.	Conceptual Understanding	05	
2.	Conduction of the exercise	05	
3.	Result Interpretation & Analysis	05	
4.	Viva Voce	05	
	Total	20	
	Signature with Date		

Formulation and Solution to Linear programming problems – Minimization

Exercise 3

Problem 1:

A leather goods company manufactures two types of leather soccer balls X and Y. Each type of ball requires work by two types of employees – semiskilled and skilled. Basically, the semiskilled employees use machines, while the skilled employees stitch the balls. The available time (per week) for each type of employees and the time requirement for manufacturing each type of balls are given below:

Type of Employees	Manufacturing Time Requirement (hr)		Time available (hr/week)
	Ball X	Ball Y	
Semi-skilled	2	3	80
Skilled	4	6	150

The cost of an hour of semiskilled labour is Rs. 5.50 and that of an hour of skilled labour is Rs. 8.50. To meet the weekly demand requirements, at least 15 balls of type X and at least 10 balls of the type Y must be manufactured. Formulate this problem as an LP model so as to minimize cost of production.

Formulation:

Decision Variables:

Let x_1 number of balls of type x to be produced.

Let x_2 number of balls of type y to be produced.

Objective Function:

$$\text{Min } Z = (2x_1 + 3x_2)\,5.5 + (4x_1 + 6x_2)\,8.5$$
$$= 11x_1 + 16.5x_2 + 34x_1 + 51x_2$$
$$= 45x_1 + 67.5x_2$$

Constraints:

Semi Skilled $\quad 2x_1 + 3x_2 \leq 80$

Skilled $\quad 4x_1 + 6x_2 \leq 150$

$x_1 \geq 15 \qquad x_2 \geq 10$

Non-negative constraints $\quad x_1, x_2 \geq 0$

Inference: Min $Z = ₹\,1350$

when $x_1 = 15$, $x_2 = 10$

Problem 2:

Evening shift resident doctors in a Government hospital work five consecutive days and have two consecutive days off. Their five days of work can start on any day of the week and the schedule rotates indefinitely. The hospital requires the following minimum number of doctors working:

Sun	Mon	Tues	Wed	Thurs	Fri	Sat
35	55	60	50	60	50	45

No more than 40 doctors can start their five consecutive working days on the same day. Formulate the general Linear Programming Model to minimize the number of doctors employed by the hospital.

Formulation:

Decision Variables:

Let x_1 be number of doctors on sunday, x_2 on monday x_3 on tuesday, x_4 on wednesday, x_5 on thursday, x_6 on friday, x_7 on saturday

Objective Function: To minimise the number of doctors

$$\text{Min } Z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7$$

To satisfy daily number of doctors

Constraints:

$$x_1 + x_4 + x_5 + x_6 + x_7 \geq 35$$
$$x_2 + x_5 + x_6 + x_7 + x_1 \geq 55$$
$$x_3 + x_6 + x_7 + x_1 + x_2 \geq 60$$
$$x_4 + x_2 + x_1 + x_2 + x_3 \geq 50$$
$$x_5 + x_1 + x_2 + x_3 + x_4 \geq 60$$
$$x_6 + x_2 + x_3 + x_4 + x_5 \geq 50$$
$$x_7 + x_3 + x_4 + x_5 + x_6 \geq 45$$

No more than 40 doctors can start their 5 consecutive days

$$x_1 \leq 40,\ x_2 \leq 40,\ x_3 \leq 40,$$
$$x_4 \leq 40,\ x_5 \leq 40,\ x_6 \leq 40,$$
$$x_7 \leq 40$$

Non-negative constraints

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0$$

Inference:

Min $Z = 73.33 \approx 73$ Doctors

$x_1 = 20$, $x_2 = 8.33$, $x_3 = 18.3$, $x_4 = 0$,

$x_5 = 13.33$, $x_6 = 10$, $x_7 = 3.33$

Problem 3:

A paint manufacturing company manufactures paints at two plants. Firm orders have been received from three large contractors. The firm has determined that the following shipping cost data is appropriate for these contractors with respect to its two plants.

Contractor	Order size (gallon)	Shipping cost/gallon(Rs)	
		From plant 1	From plant 2
A	750	1.80	2.00
B	1500	2.60	2.20
C	1500	2.10	2.25

Each gallon of paint must be blended and tinted. The company's costs with respect to these two operations at each of the two plants are as follows:

Plant/Operation		Hours required per gallon	Cost/hours (Rs)	Hours available
Blending	Plant1	0.10	3.80	300
Tinting		0.25	3.20	360
Blending	Plant 2	0.15	4.00	600
Tinting		0.20	3.10	720

Formulate this problem as a linear programming model.

Formulation:

Decision Variables:

Let X_{1A}, X_{1B} and X_{1C} be the gallons of paint shipped to plant 1 to contractor A, B and C respectively.

Let X_{2A}, X_{2B} and X_{2C} be the gallons of paint shipped to plant 2 to contractor A, B and C respectively.

Objective Function:

$$Z_{min} = 1.80X_{1A} + 2.60X_{1B} + 2.10X_{1C} + 2.00X_{2A} + 2.20X_{2B} + 2.25X_{2C} +$$
$$3.80(0.1X_{1A}+0.1X_{1B}+0.1X_{1C}) + 3.20(0.25X_{1A} + 0.25X_{1B} + 0.25X_{1C}) +$$
$$4.00(0.15X_{1A} + 0.15X_{2B} + 0.15_{2C}) + 3.10(0.20X_{2A} + 0.20X_{2B} + 0.20X_{2C})$$
$$= \text{Shipping Cost} + \text{Operation Cost}$$

Constraints:

Contractor A $X_{1A} + X_{2A} = 750$

Contractor B $X_{1B} + X_{2B} = 1500$

Contractor C $X_{1C} + X_{2C} = 1500$

$0.1(X_{1A} + X_{1B} + X_{1C}) \leq 300$

$0.25(X_{1A} + X_{1B} + X_{1C}) \leq 360$

$0.15(X_{2A} + X_{2B} + X_{2C}) \leq 600$

$0.20(X_{2A} + X_{2B} + X_{2C}) \leq 720$

Non-negativity Constraint

$X_{1A}, X_{1B}, X_{1C}, X_{2A}, X_{2B}, X_{2C} \geq 0$

Inference:

Min Z = ₹ 12438.9

$x_{1A} = 750$, $x_{1B} = 0$, $x_{1C} = 690$

$x_{2A} = 0$, $x_{2B} = 1500$, $x_{2C} = 810$

Sl. No	Particulars	Max Marks	Marks Obtained
1.	Conceptual Understanding	05	
2.	Conduction of the exercise	05	
3.	Result Interpretation & Analysis	05	
4.	Viva Voce	05	
	Total	20	
	Signature with Date		

Sensitivity Analysis – Maximization

Exercise 4

Problem 1:

A market gardener grows red and white rose bushes. Each red costs £8 per year to grow and each white £2 per year. The red and white bushes require an area of $5dm^2$ and $4dm^2$ per bush respectively. The labour needed per year for a red bush is 1 – man hour, whereas for a white it is 5 man hours. The reds yield a profit of £2, and the whites £3. The total land available is at most $6,100dm^2$, and the total available finance is £8,000. The labour available is at most 5,000 man hours per year.

1. How many of each type of rose bush should be planted to maximize the profit? (Use Excel solver)
2. Which resources (land, finance, labour) are used up fully and which resources are abundant?
3. How much extra land would be required to increase the total profit if all the finances are to be used up? What is the new optimum profit?
4. If the same combination of reds and whites as in (1) is to be the optimum, what is the range of profit contribution from reds and whites towards total profit?
5. Manually plot the constraints and objective function lines in a graph sheet.

Formulation:

Decision Variables:

Let x_1 be the number of red bushes

Let x_2 be the number of white bushes

Objective Function:

The red bush yield a profit of £2 and white bushes yield a profit of £3. Max $Z = 2x_1 + 3x_2$

Constraints:

Cost $8x_1 + 2x_2 \leq 8000$

Area $5x_1 + 4x_2 \leq 6100$

Man hours $x_1 + 5x_2 \leq 5000$

Non-negativity Constraints

$x_1, x_2 \geq 0$

Solution using Microsoft excel:

1) $Z = £3700$

$x_1 = 500$

$x_2 = 900$

Inference:

2) Area and man hours are hours used fully finance is in abundance

3) Extra area required = 1215.79 dm^3

new profit = £4105.263

4) With respect to 1

profit on red = 500 × 2 = £1000

profit on white = 900 × 3 = £2700

Problem 2:

A factory manufactures three products A, B and C for which the data is given in the table below. Find the optimal product mix if the profit/unit is Rs. 32, Rs. 30 and Rs. 40 for product A, B and C respectively.

	Product			Available Resources
	A	B	C	
Material required (kg/unit)	5	4	3	2,500 kg
Machine hours required/unit	2	3	1	1,275 hours
Labour hours require/unit	3	2	4	2,100 hours

a)Find the optimal solution if machine hours available become 1,350 instead of 1,275.
b)Find the optimal solution if labour hours become 2,000 instead of 2,100.
c)Find the optimal solution if 10 units of product *A* are to be produced.

Formulation:

Decision Variables:

Let x_1 be the no. of units of A manufactured.

Let x_2 be the no. of units of B manufactured.

Let x_3 be the no. of units of C manufactured

Objective Function: The profit on each unit of A, B and C is Rs. 32, Rs. 30 and Rs. 40 respectively.

$$\text{Max } Z = 32x_1 + 30x_2 + 40x_3$$

Constraints:

Material required $- 5x_1 + 4x_2 + 3x_3 \leq 2500$

Machine hours $- 2x_1 + 3x_1 + x_3 \leq 1275$

Labour hours $- 3x_1 + 2x_2 + 4x_3 \leq 2100$

Solution using Microsoft excel:

c) $Z = ₹24300$

$x_1 = 0$

$x_2 = 330$

$x_3 = 360$

b) $Z = ₹23100$

$x_1 = 0$

$x_2 = 310$

$x_3 = 345$

c) $Z = ₹3940$

$x_1 = 10$

$x_2 = 295$

$x_3 = 370$

Inference: Max Z = ₹ 24000

when $x_1 = 0$, $x_2 = 300$, $x_3 = 375$

- The optimal values of decision variable are obtained.
- Maximum profit is achieved by satisfying all constraints.

Sl. No	Particulars	Max Marks	Marks Obtained
1.	Conceptual Understanding	05	
		05	

Sensitivity Analysis – Minimization

Exercise 5

Problem 1:

A pig farmer is attempting to analyse his feeding operation. The minimum daily requirement of the three nutritional elements for the pigs and the number of units of each of these nutritional elements in two feeds is given in the following table:

Required Nutritional Element	Units of Nutritional Elements (in kg)		Minimum Requirement
	Food 1	Food 2	
A	20	30	200
B	40	25	350
C	30	45	430
Cost per kg (Rs.)	5	3	

(a) Formulate and solve this problem as an LP model.

(b) Assume that the farmer can purchase a third feed at a cost of Rs. 2 per kg, which will provide 35 units of nutrient A, 30 units of nutrient B and 50 units of nutrient C. Would this change the optimal mix of feeds? If yes, how?

Formulation:

Decision Variables:

Let x_1 be the production volume of food 1

Let x_2 be the production volume of food 2

Objective Function:

Cost of food 1 is ₹5 and cost of food 2 is ₹3,

$$\text{Min } Z = 5x_1 + 3x_2$$

Constraints:

$$20x_1 + 30x_2 \geq 200$$
$$40x_1 + 25x_2 \geq 350$$
$$30x_1 + 45x_2 \geq 430$$

Non-negativity Constraint

$$x_1, x_2 \geq 0$$

Solution using Microsoft excel:

(a) $Z = ₹42$

$x_1 = 0$, $x_2 = 14$

(b) Yes

$Z = ₹23.33$

$x_1 = 0$, $x_2 = 0$, $x_3 = 11.67$

Inference:

- The optimal values of decison variables are obtained.
- Minimum cost is achieved while satisfying all constraints.
- The solution gives the most economical plan.

Problem 2:

A manufacturing plant needs to mix two raw materials to meet production standards for strength, flexibility, and durability. The composition per kg of each material and the required minimum values are:

Property	Material 1 (kg)	Material 2 (kg)	Minimum Requirement
Strength	50	30	400
Flexibility	20	40	250
Durability	35	45	300
Cost per kg (Rs.)	**8**	**6**	

a) Develop an LP model to minimize cost while meeting the requirements.

b) If a third material is available at Rs. 5 per kg with 40 kg strength, 35 kg flexibility, and 50 kg durability, analyze its impact on cost.

Formulation:

Decision Variables:

Let x_1 be amount of material 1

Let x_2 be amount of material 2

Objective Function:

Cost of material 1 and material 2 per kg is ₹8 and ₹6

$$\text{Min } Z = 8x_1 + 6x_2$$

Constraints:

Strength $-\ 50x_1 + 30x_2 \geq 400$

Flexibility $-\ 20x_1 + 40x_2 \geq 250$

Durability $-\ 35x_1 + 45x_2 \geq 300$

Solution using Microsoft excel:

(a) $Z = ₹67.857$

$x_1 = 8.07$

$x_2 = 3.214$

(b) $Z = ₹50$

$x_1 = 0$

$x_2 = 0$

$x_3 = 10$

Inference:

- Optimal Values are Obtained.

Sl. No	Particulars	Max Marks	Marks Obtained
1.	Conceptual Understanding	05	
2.	Conduction of the exercise	05	
3.	Result Interpretation & Analysis	05	
4.	Viva Voce	05	
	Total	20	
	Signature with Date		

Problem 1:

A product is manufactured at four factories A, B, C and D. Their unit production costs are Rs. 2, Rs. 3, Re. 1 and Rs. 5, respectively. Their production capacities are 50, 70, 30 and 50 units, respectively. These factories supply the product to four stores, demands of which are 25, 35, 105 and 20 units respectively. Unit transportation cost in rupees from each factory to each store is given in the table below.

Factories	Stores I	II	III	IV
A	2	4	6	11
B	10	8	7	5
C	13	3	9	12
D	4	6	8	3

Determine the extent of deliveries from each of the factories to each of the stores, so that the total production and transportation cost is the minimum.

FROM \ TO	Store I	Store II	Store III	Store IV	Supply
A	2	4	6	11	50
B	10	8	7	5	70
C	13	3	9	12	30
D	4	6	8	3	50
Demand	25	35	105	20	

Solution using Microsoft excel:

	Store I	Store II	Store III	Store IV	SUPPLY
A	10	5	35	0	50
B	0	0	70	0	70
C	0	30	0	0	30
D	15	0	0	20	35
DEMAND	25	35	105	20	

Inference:

$Z_{min} = ₹ 1465$

- Total transportation cost is minimised
- Resources are allocated efficiently to meet all requirements.

Problem 2:

Consider the problem of scheduling the weekly production of certain items for the next four weeks. The production cost of the item is Rs 10 for the first two weeks and Rs 15 for the last two weeks. The weekly demands are 300, 700, 900 and 800, which must be met. The plant can produce a maximum of 700 units per week. In addition, the company can use overtime during the second and third week. This increases the weekly production by an additional 200 units, but the production cost also increases by Rs 5. Excess production can be stored at a unit cost of Rs 3 per week. How should the production be scheduled so as to minimize the total cost?

COST	W1	W2	W3	W4	Supply
W1	10	13	16	19	700
W2 reg	-	10	13	16	700
W2 reg	-	15	18	21	200
W3 reg	-	-	15	18	700
W4 OT	-	-	20	23	200
W4 reg	-	-	-	15	700

Solution using Microsoft excel:

Z = 36100

Inference:

- The optimal allocation is obtained.
- Total transportation cost is minimised
- Resources are allocated efficiently to meet all requirements.

Problem 3:

ABC company wishes to develop a monthly production schedule for the next three months. Depending upon the sales commitments, the company can either keep the production constant, allowing fluctuations in inventory, or inventories can be maintained at a constant level, with fluctuating production. Fluctuating production necessitates working overtime, the cost of which is estimated to be double the normal production cost of Rs. 12 per unit. Fluctuating inventories result in inventory carrying cost of Rs. 2 per unit. If the company fails to fulfill its sales commitment, it incurs a shortage cost of Rs. 4 per unit per month. The production capacities for the next three months are shown in the following table:

	Production Capacity		
Month	Regular	Overtime	Sales
M_1	50	30	60
M_2	50	0	120
M_3	60	50	40

Determine the optimal production schedule.

COST	MONTH 1	MONTH 2	MONTH 3	SUPPLY
M_1 Regular	12	14	16	50
M_1 Overtime	24	26	28	30
M_2 Regular	16	12	14	50
M_2 Overtime	28	24	26	0
M_3 Regular	20	16	12	60
M_3 Overtime	32	28	24	50
DEMAND	60	120	40	

Solution using Microsoft excel:

	MONTH 1	MONTH 2	MONTH 3	SUPPLY
M_1	30	20	0	50
M_1	30	0	0	30
M_2	0	50	0	50
M_2	0	0	0	0
M_3	0	20	0	60
M_3	0	30	40	30
DEMAND	60	120	0	

Inference:

Min Z = 3600

- The production problem was formulated and solved as transportation problem.
- Transportation models can act effectively to support production planning decisions.

Design of a spread sheet for Assignment Problem

Exercise 7

Problem 1:

The hospital administrator at St. Charles General Hospital must appoint head nurses to four newly established departments: urology, cardiology, orthopedic, and obstetrics. In anticipation of this staffing problem, she has hired four nurses: Hawkins, Condriac, Bardot, and Hoolihan. Each nurse has been ranked on a cost scale of 0 to 100 for each department; a 0 implies that she would be perfectly suited for the task; a 100 implies that the nurse is not at all suited to head the unit. The table below represents all possible assignments and nurse ratings. Which nurse should be assigned to which unit?

	Department			
Nurse	**Urology**	**Cardiology**	**Orthopaedics**	**Obstetrics**
Hawkins	20	18	15	75
Condriac	32	48	23	38
Bardot	51	36	24	36
Hoolihan	25	38	55	12

COST	Urology	Cardiology	Orthopaedics	Obstetrics	SUPPLY
Hawkins	0	1	0	0	1
Condriac	1	0	0	0	1
Bardot	0	0	1	0	1
Hoolihan	0	0	0	1	1
DEMAND	1	1	1	1	

$Z_{min} = 86$

Problem 2:

A company has four managers (Anderson, Blake, Carter, and Dawson) and four new projects (AI Development, Market Expansion, Infrastructure Upgrade, and Cybersecurity). Each manager is rated based on their suitability for each project on a cost scale from 0 to 100. The company must assign each manager to one project to minimize the total cost.

Manager	AI Development	Market Expansion	Infrastructure Upgrade	Cybersecurity
Anderson	78	85	90	82
Blake	92	88	97	85
Carter	85	80	95	83
Dawson	91	87	92	90

	AI DEV	MARKET EX.	INFRA UPG.	CYBER SEC.	SUPPLY
Anderson	1	0	0	0	1
Blake	0	0	0	1	1
Carter	0	1	0	0	1
Dawson	0	0	1	0	1
DEMAND	1	1	1	1	1

Min Z = ₹ 335

Sl. No	Particulars	Max Marks	Marks Obtained
1.	Conceptual Understanding	05	
2.	Conduction of the exercise	05	
3.	Result Interpretation & Analysis	05	
4.	Viva Voce	05	
	Total	20	
	Signature with Date		

Project Network Analysis

Exercise 8

Problem 1:

A project consisting of the following activities has precedence relation and time duration as follows:

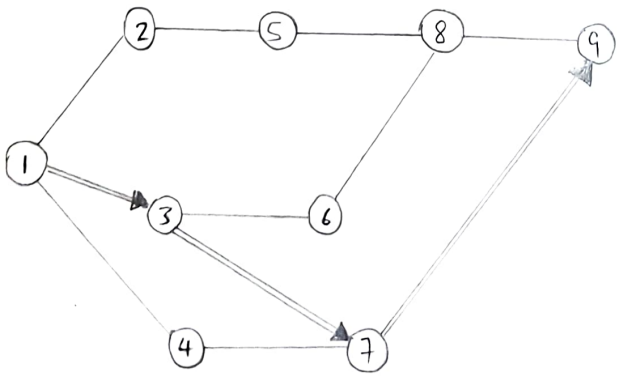
Activity	A	B	C	D	E	F	G	H	I	J	K
Predecessor	-	-	-	A	B	B	C	D	E	H,I	F,G
Duration	4	5	2	1	6	9	3	7	1	4	8

i. Construct network diagrams.
ii. Identify critical path, compute project duration, total and free floats.
iii. Construct a schedule diagram for the project.
iv. If instead of firm duration estimates of activities, 3 estimates (optimistic, most likely & pessimistic) are given as below, how does the critical path and project duration change? What is the variance in project duration?
v. In the above case, what is the probability of the project being completed within 3 days of delay? Two days earlier?

Activity	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**
Optimistic	2	3	3	2	2	3	1	4	1	3	2
Most likely	2	4	7	3	7	3	4	6	2	7	5
Pessimistic	6	5	10	6	8	9	5	7	4	8	7

Solution:

Network Diagram:



Critical Path & Project Duration:

Critical Path

1 - 3 - 7 - 9

Duration

= 5 + 9 + 8

= 22

Total float & Free Float:

Activity	A	B	C	D	E		

Problem 2:

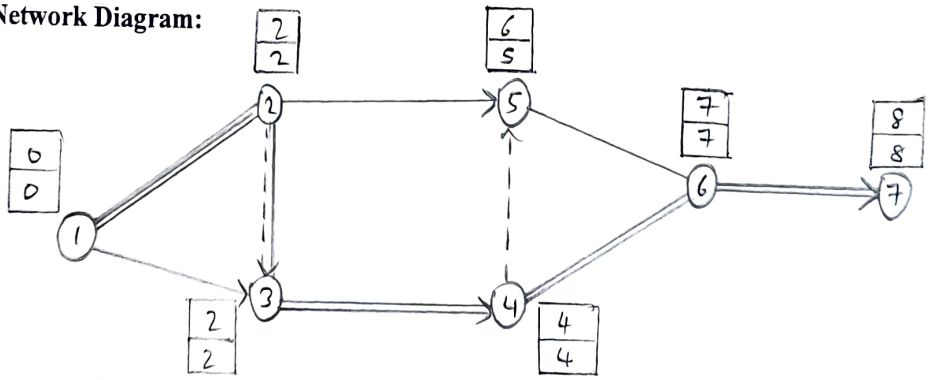
An architect has been awarded a contract to prepare plans for an urban renewal project. The job consists of the following activities and their estimated times:

Activity	Description	Immediate Predecessors	Time (days)
A	Prepare prelimnary sketches	--	2
B	Outline specifications	--	1
C	Prepare drawings	A	3
D	Write specificatons	A, B	2
E	Run off prints	C, D	1
F	Have specification	B, D	3
G	Assemble bid packages	E, F	1

(a)Draw the network diagram of activities for the project.
(b)Indicate the critical path, and calculate the total float and free float for each activity.

Solution:

Network Diagram:



Critical Path : 1-2-3-4-6-7

Critical Path & Project Duration:

CPM : 1 - 2 - 3 - 4 - 6 - 7

Project Duration : 2 + 2 + 3 + 1 = 8 Day

ACTIVITY	L_j	E_j	E_{ij}	F_{ij}	TF	FF
A	2	2	0	2	0	0
B	2	2	0	1	1	1
C	6	5	2	3	1	0
D	4	4	2	2	0	0
E	7	7	5	1	1	1
F	7	7	4	3	0	0
G	8	8	7	1	0	0

Total float & Free Float:

Activity	A	B	C	D	E	F	G
Total Float	0	1	1	0	1	0	0
Free Float	0	1	0	0	1	0	0

Sl. No	Particulars	Max Marks	Marks Obtained
1.	Conceptual Understanding	05	
2.	Conduction of the exercise	05	
3.	Result Interpretation & Analysis	05	
4.	Viva Voce	05	
	Total	20	
	Signature with Date		

Game Theory

Exercise 09

Problem 1:

A company is currently involved in negotiations with its union on the upcoming wage contract. Positive signs in table represent wage increase while negative sign represents wage reduction. What are the optimal strategies for the company as well as the union? What is the game value?

Conditional costs to the company (Rs. in Lakhs)

		Union Strategies			
		U_1	U_2	U_3	U_4
Company Strategies	C_1	0.25	0.27	0.35	-0.02
	C_2	0.20	0.06	0.08	0.08
	C_3	0.14	0.12	0.05	0.03
	C_4	0.30	0.14	0.19	0.00

Solution:

	V	x_1	x_2	x_3	x_4		
Maximize	1	0	0	0	0	$\leq$	
	1	-0.25	-0.20	-0.14	-0.30	$\leq$	0
	1	-0.27	0.04	-0.12	0.14	$\leq$	0
	1	-0.35	-0.06	-0.05	-0.09	$\leq$	0
	1	0.02	-0.08	-0.03	0.00	$\leq$	0
	1	1.00	1.00	1.00	1.0	$\leq$	1
	y	n	n	n	n		

Problem 2:

Two breakfast food manufacturers, ABC and XYZ are competing for an increased market share. The payoff matrix, shown in the following table, describes the increase in market share for ABC and decrease in market share of XYZ.

ABC	XYZ			
	Give Coupons	Decrease Price	Maintain Present Strategy	Increase Advertising
Give Coupons	2	-2	4	1
Decrease Price	6	1	12	3
Maintain Present Strategy	-3	2	0	6
Increase Advertising	2	-3	7	1

Determine the optimal strategies for both the manufacturers and the value of the game.

	V	x_1	x_2	x_3	x_4		
MAX	1	0	0	0	0	<=	0
	1	-2	-6	3	-2	<=	0
	1	2	-1	-2	3	<=	0
	1	-9	-10	0	-7	<=	0
	1	1	-3	-6	1	<=	1
	0	1	1	1	1	=	

Solution:

Optimal Strategy for ABC = $(0, 0.5, 0.5, 0)$

Optimal Strategy for XYZ = $(0.1, 0.9, 0, 0)$

Value of the Game $\Rightarrow V = 1.5$

Sl. No	Particulars	Max Marks	Marks Obtained
1.	Conceptual Understanding	05	
2.	Conduction of the exercise	05	
3.	Result Interpretation & Analysis	05	
4.	Viva Voce	05	
	Total	20	
	Signature with Date		